

The Fixed-Point Converse in Remark 5.3 Is False

A four-element counterexample with nontrivial character

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9 August 2026

Status. Candidate counterexample, likely valid, pending human review. The construction answers the literal two-sided wording of Remark 5.3. It does not settle the repaired one-sided question obtained by replacing “ φ -amenable” with “left φ -amenable.”

1 Source question

Jabbari, Ebadian, and Eshaghi Gordji introduce φ -amenability for topological semigroups in [1]. For a continuous character $\varphi : S \rightarrow \mathbb{T}$, they write

$$P(\varphi) = \{s \in S : \varphi(s) = 1\}.$$

Corollary 5.2(ii) is the assertion that every continuous affine action of $P(\varphi)$ on a nonempty compact convex subset of a locally convex space has a common fixed point. Remark 5.3 asks whether this condition implies that S is φ -amenable. The source defines φ -amenability to mean *both* left and right φ -amenability.

2 Idea of the proof

The fixed-point condition is one-sided. A right-zero semigroup $\mathcal{R} = \{0, 1\}$, whose product is $ij = j$, has exactly the required affine fixed-point property: a fixed point of either one of the action maps is automatically fixed by both. On the opposite side, however, right translations by the two idempotents collapse $\mathcal{R}$ onto two different fibers. No normalized invariant functional can survive both collapses.

To make the character nontrivial, attach a cyclic sign coordinate. Thus $S = C_2 \times \mathcal{R}$ and φ is the sign character on C_2 . Its kernel $P(\varphi)$ is precisely the right-zero factor. A short quotient-module argument derives the needed invariant functional directly from the definition of right φ -amenability; two indicator functions then give the contradiction.

3 A necessary functional for right φ -amenability

For $s \in S$, let R_s denote right translation on $C(S)$,

$$(R_s f)(t) = f(ts).$$

The next finite-dimensional lemma is included so that the counterexample does not rely on a separate mean-characterization theorem from the source.

fixed point Theorem for topological semigroups is explained and investigated by Argabright and Mitchell works. Namioka in [19] showed that $LUC(S)$ is a translation-invariant subspace of $C_r(S)$ [19] Lemma 2] and similarly, one can see that $RUC(S)$ is a translation-invariant subspace of $C_r(S)$. We recall the following result:

Theorem 5.1. [16] Theorem 1] *Let S be a topological semigroup. Then the following assertions equivalent:*

- (i) $RUC(S)$ has a multiplicative left invariant mean.
- (ii) whenever S acts on a compact Hausdorg space Y , where the map $S \times Y \longrightarrow Y$ is jointly continuous, then Y contains a common fixed point of S .

Let S be a topological semigroup, $\varphi \in \Delta_S(S)$ and $P(\varphi)$ be as defined in the previous section. Clearly, $P(\varphi)$ is a topological subsemigroup of S such $\varphi|_{P(\varphi)} = 1$. Then by replacing S by $P(\varphi)$ in Theorem 5.1 we have the following result that we give its proof because we use some of the obtained results in the proof, for the last result of this paper:

Corollary 5.2. *Let S be a topological semigroup and $\varphi \in \Delta_S(S)$. Then the following assertions equivalent:*

- (i) *There is a bounded linear functional $m \in RUC(P(\varphi))^*$ such that $m(\varphi) = 1$ and $m(f \cdot s) = m(f)$, for all $f \in RUC(P(\varphi))$ and $s \in P(\varphi)$.*
- (ii) *Y contains a common fixed point of $P(\varphi)$ for continuous affine actions of $P(\varphi)$ on compact convex sets of a locally convex linear topological space.*

Figure 1: The complete fixed-point condition in Corollary 5.2, source page 17.

Lemma 1. *Let S be a finite discrete semigroup and let $\varphi : S \rightarrow \mathbb{T}$ be a character. If S is right φ -amenable in the sense of [1], then there is $m \in C(S)^*$ such that*

$$m(\varphi) = 1, \quad m(R_s f) = \varphi(s)m(f) \quad (s \in S, f \in C(S)). \quad (1)$$

Proof. Put $E = C(S)$ and make it an S -bimodule by

$$(s \cdot f)(t) = f(ts), \quad (f \cdot s)(t) = \varphi(s)f(t).$$

The line $\mathbb{C}\varphi$ is an S -submodule, because $s \cdot \varphi = \varphi \cdot s = \varphi(s)\varphi$. Hence $X = E/\mathbb{C}\varphi$ is a Banach S -bimodule whose right action is $x \cdot s = \varphi(s)x$, exactly as required in the definition of right φ -amenability. Identify

$$X^* = Y = \{F \in E^* : F(\varphi) = 0\}.$$

Choose $F_0 \in E^*$ with $F_0(\varphi) = 1$. With the usual dual module actions, define

$$D(s) = s \cdot F_0 - F_0 \cdot s.$$

This is an inner derivation in E^* and it takes values in Y , since its two terms have the same value $\varphi(s)$ on φ . Because S is finite and discrete, D is bounded and continuous. Right φ -amenability therefore supplies $F \in Y$ such that

$$D(s) = s \cdot F - F \cdot s \quad (s \in S).$$

Set $m = F_0 - F$. Then $m(\varphi) = 1$ and $s \cdot m = m \cdot s$. Evaluating this identity at $f \in E$ gives

$$\varphi(s)m(f) = m(f \cdot s) = (s \cdot m)(f) = (m \cdot s)(f) = m(s \cdot f) = m(R_s f),$$

which is (1). □

for all $h \in A(Y)$ and $s \in P(\varphi)$. This shows that $sy' = y'$.

(ii) $\implies$ (i) Set $X = RUC(P(\varphi))^*$ with w^* -topology and suppose that Y is the compact convex set of all bounded linear functionals on $RUC(P(\varphi))$ such that, for each $m \in Y$, $m(\varphi) = 1$. Define the affine action $T : P(\varphi) \times Y \longrightarrow Y$ by $T_s\mu = l_s^*\mu$, for all $s \in P(\varphi)$ and $\mu \in Y$, where l_s^* is the adjoint of l_s . It is easy to check that T is a uniformly continuous affine action on Y . Thus, (ii) implies that there exists $m \in Y$ that is fixed under the affine action T of $P(\varphi)$ such that $l_s^*m = s \cdot m = m$. This means that $s \cdot m(f) = m(f \cdot s) = m(f)$. Thus, (ii) implies (i). $\square$

Remark 5.3. *Note that if a topological semigroup S is φ -amenable, where $\varphi \in \Delta_S(S)$, then the condition (ii) holds by the same reasons in the proof of the above Corollary, but, we do not know the converse part holds or not?*

Similar to Corollary 5.2, one can consider the conditions weakly right uniformly continuous functions on S instead of right uniformly continuous functions, we do not consider this case in this paper.

Suppose that E is a separated locally convex space and X is a subset of E containing an n -dimensional subspace. Let S be a topological semigroup and $\mathcal{T} = \{T_s : s \in S\}$

Figure 2: Remark 5.3 and its immediate context, source page 18.

4 Counterexample

Theorem 1. *The converse asked for in Remark 5.3 of [1] is false, even when S is a four-element discrete semigroup and φ is nontrivial.*

Proof. Let $C_2 = \{0, 1\}$ with addition modulo two and let $\mathcal{R} = \{0, 1\}$ be the right-zero semigroup, so $ij = j$. On

$$S = C_2 \times \mathcal{R}$$

the direct-product multiplication is

$$(\varepsilon, i)(\delta, j) = (\varepsilon + \delta \pmod{2}, j). \quad (2)$$

Give S the discrete topology and define

$$\varphi(\varepsilon, i) = (-1)^\varepsilon.$$

Associativity follows from the direct-product construction, and (2) immediately gives $\varphi(xy) = \varphi(x)\varphi(y)$. Thus φ is a continuous nontrivial character. Moreover,

$$P(\varphi) = \{p_0, p_1\}, \quad p_i = (0, i), \quad p_i p_j = p_j.$$

We first verify Corollary 5.2(ii). Let T_p be any continuous affine action of $P(\varphi)$ on a nonempty compact convex set K in a locally convex space. By the Schauder–Tychonoff fixed-point theorem [2], the map T_{p_0} has a fixed point $y \in K$. For either $p \in P(\varphi)$,

$$T_p y = T_p T_{p_0} y = T_{pp_0} y = T_{p_0} y = y,$$

because $pp_0 = p_0$. Hence y is a common fixed point, and condition (ii) holds.

It remains to show that S is not φ -amenable. Suppose instead that S were right φ -amenable. Lemma 1 would give a functional m satisfying (1). For $i = 0, 1$, let

$$u_i(\varepsilon, j) = \begin{cases} 1, & j = i, \\ 0, & j \neq i. \end{cases}$$

Right multiplication by p_i sends (ε, j) to (ε, i) . Consequently

$$R_{p_i}(\varphi u_{1-i}) = 0.$$

Since $\varphi(p_i) = 1$, equation (1) implies

$$m(\varphi u_{1-i}) = m(R_{p_i}(\varphi u_{1-i})) = 0.$$

Using $i = 0$ and $i = 1$ yields $m(\varphi u_0) = m(\varphi u_1) = 0$. But $u_0 + u_1 = 1$, so

$$1 = m(\varphi) = m(\varphi u_0) + m(\varphi u_1) = 0,$$

a contradiction. Thus S is not right φ -amenable and therefore is not φ -amenable. This disproves the converse. $\square$

Remark 1 (A two-element version). *If the trivial character is allowed, take $S = \mathcal{R}$ itself and $\varphi \equiv 1$. The same fixed-point proof applies. If a functional from Lemma 1 existed, right translation by i would annihilate u_{1-i} , forcing $m(u_0) = m(u_1) = 0$ and contradicting $m(1) = 1$. Thus two elements already suffice for the literal question; Theorem 1 shows that triviality of the character is not the issue.*

5 Verification, scope, and novelty

The proof is exact and finite-dimensional. The accompanying verifier checks all 64 associativity triples, all 16 character identities, the right-zero kernel, and the complete invariant-functional system over the rational numbers. That system has coefficient rank 4 and augmented rank 5.

The left/right distinction is essential. Corollary 5.2(ii) is a one-sided fixed-point property of $P(\varphi)$, whereas the source’s unqualified term “ φ -amenable” is explicitly two-sided. The theorem completely answers Remark 5.3 as written. It does not resolve the repaired question whether condition (ii) forces *left* φ -amenability.

The run indexes contained no matching result, attempt, or proof-gap packet. Bounded web/arXiv searches on 9 August 2026 used the exact Remark 5.3 wording, the title, DOI and authors, and combinations of $P(\varphi)$, fixed points, right-zero semigroups, and character amenability. They found the source and bibliographic records but no later answer, correction, or matching counterexample. This is not an exhaustive novelty claim.

References

- [1] A. Jabbari, A. Ebadian, and M. Eshaghi Gordji, *A Weak Form of Amenability of Topological Semigroups and its Applications in Ergodic and Fixed Point Theories*, arXiv:2005.08089 (2020); Collect. Math. **74** (2023), 149–171, doi:10.1007/s13348-021-00340-7.
- [2] A. Tychonoff, *Ein Fixpunktsatz*, Math. Ann. **111** (1935), 767–776.